

6	(a)	uniform electric (E) and magnetic fields (B) normal to each other AND charged particle enters region normal to both fields correct B direction w.r.t. E for zero deflection For no deflection, $v = E/B$	B1 B1 B1
	(b)	magnetic force provides the centripetal force $BQv = mv^2/r$ $r = mv / BQ$	M1 A1
	(c)	curve of larger radius starting from common entry point into the field	A1
	(d)	$r = mv / BQ$, so B proportional to m $B = (0.680) \times (87/86) = 0.6879 \text{ T}$ $\Delta B = 7.9 \times 10^{-3} \text{ T}$	C1 A1

7	(a)		
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